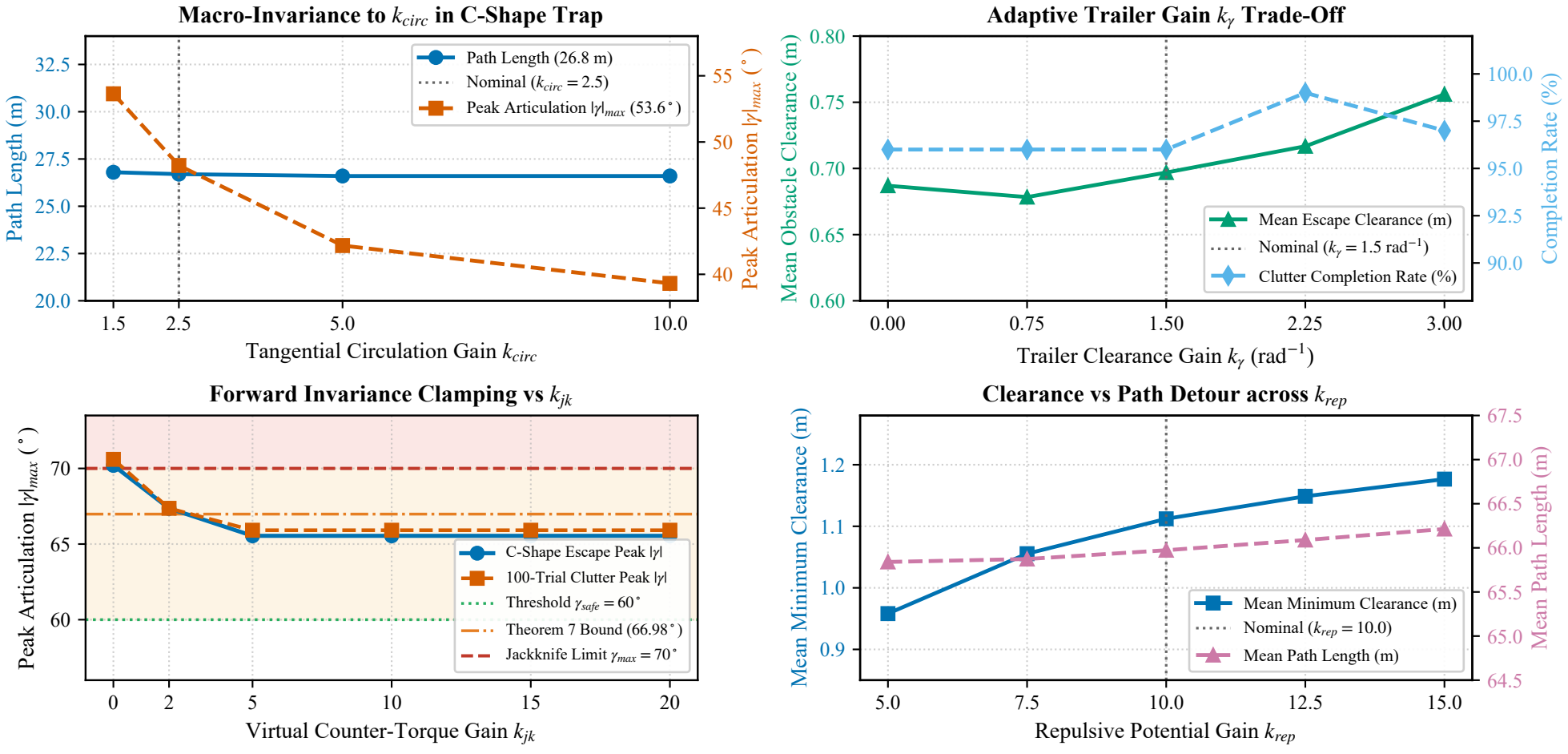

Macro-Invariance to k_{circ} in C-Shape Trap
Path Length (m)
Peak Articulation $|\gamma|_{max}$ (°)
Path Length (26.8 m)
Nominal ($k_{circ} = 2.5$)
Peak Articulation $|\gamma|_{max}$ (53.6°)
Tangential Circulation Gain k_{circ}
Adaptive Trailer Gain k_γ Trade-Off
Mean Obstacle Clearance (m)
Completion Rate (%)
Mean Escape Clearance (m)
Nominal ($k_\gamma = 1.5$ rad^{-1})
Clutter Completion Rate (%)
Trailer Clearance Gain k_γ (rad^{-1})
Forward Invariance Clamping vs k_{jk}
Peak Articulation $|\gamma|_{max}$ (°)
C-Shape Escape Peak $|\gamma|$
100-Trial Clutter Peak $|\gamma|$
Threshold $\gamma_{safe} = 60°$
Theorem 7 Bound (66.98°)
Jackknife Limit $\gamma_{max} = 70°$
Virtual Counter-Torque Gain k_{jk}
Clearance vs Path Detour across k_{rep}
Mean Minimum Clearance (m)
Mean Path Length (m)
Mean Minimum Clearance (m)
Nominal ($k_{rep} = 10.0$)
Mean Path Length (m)
Repulsive Potential Gain k_{rep}